

Keren Xu

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EMPLOYMENT

Tempus AI, Inc.

Remote

Senior Bioinformatics Scientist, Bioinformatics Scientist *Mar. 2024–Present, June 2022–Mar. 2024*

- Develop bayesian probability models and loss functions for tumor purity estimation from next generation sequencing data
- Develop circular binary segmentation algorithm to identify copy number variations
- Evaluate and implement new software to improve short variant / structural variation calling and classification
- Design and conduct validation studies for Tempus first FDA-approved tumor assay (xT CDx) as part of its FDA submission
- Produce high quality and detailed documentation for all projects using git version control system
- Present project outcomes to interdisciplinary groups of scientists and engineers to translate research into clinically actionable insights

University of Southern California, Center for Genetic Epidemiology

Los Angeles, CA

Predoctoral Fellow

Aug. 2018–May 2022

- Built bioinformatics pipelines for germline/somatic variants (short variants/structural variations/copy number alterations) discovery and mutational signature analysis of over 10-terabyte human whole-genome/exome sequencing data on high-performance computing cluster
- Conducted epigenome-wide association studies, methylation quantitative trait loci analyses, causal mediation analyses, and meta-analyses to identify genetic risks that lead to leukemia by altering DNA methylation
- Conducted genome-wide association studies and construct polygenetic risk scores to assess the associations of blood cell traits with childhood ALL patient characteristics
- Detected accelerated aging in Down syndrome newborns using epigenetic age clocks derived from elastic net regression
- Assessed the impact of prenatal smoking on somatic gene deletion burden in childhood ALL patients using the polyepigenetic DNA methylation smoking score constructed by logistic lasso regression
- Held weekly lectures and office hours as a teaching assistant for 5 graduate-level courses in Biostatistics, Data analysis, and Epidemiology

Dartmouth-Hitchcock Medical Center

Lebanon, NH

Statistician

July 2017–June 2018

- Analyzed patient survey data in multiple projects using R programming
- In charge of cluster randomized trial design, survey design, IRB submission, and patient recruitment
- Collaborated with medicine fellow in writing study proposal which received \$15,000 Cardiovascular Medicine Fellowship Research Award
- Organized stakeholder group meeting including 4 clinicians and 8 patients to collect data for grant application

Columbia University Medical Center

New York, NY

Graduate Research Assistant

May 2016–July 2017

- Analyzed questionnaire data using structural equation modeling, multilevel modeling, and multinomial logistic regression in SAS and R
- Created and maintained three-year survey database in SPSS to efficiently track project outcomes

EDUCATION

University of Southern California, Keck School of Medicine

Los Angeles, CA

Doctor of Philosophy in Epidemiology, GPA: 3.92/4.00

2018–2022

- Dissertation: Genetic Epidemiological Approaches in the Study of Risk Factors for Hematologic Malignancies

Columbia University, Mailman School of Public Health

New York, NY

Master of Public Health in Epidemiology, Certificate in Advanced Epidemiology, GPA: 3.81/4.00

- Thesis: Modeling the Impacts of Numbers of Affected Relatives and Perceived Chance of Having Epilepsy-Related Mutation

East China University of Science and Technology, School of Pharmacy

Shanghai, China

Bachelor of Science in Pharmaceutical Sciences, Minor in English, GPA: 3.73/4.00

2011–2015

- Thesis: Pluripotent Stem Cells Culture and ICOSL Gene Knockout

RELEVANT SKILLS

Data Science: Prediction, Inference, Machine Learning, Visualization, Data Cleaning, Modelling.

Programming: Python, R, SQL, Linux/Unix, Bash, Git, Docker, AWS, GCP, SAS, STATA.

Epidemiology & Biostatistics: Clinical trial, Study design, Quantitative analysis, Systematic review, Meta analysis.

Bioinformatics: GATK, PLINK, Conda, Bioconductor, WDL, EWAS, GWAS, PRS.

Languages: Mandarin (Native), English (Full professional working proficiency).

CERTIFICATIONS

2023	Python 3 Programming Specialization by University of Michigan on Coursera
2023	Applied Data Science with Python Specialization by University of Michigan on Coursera
2021	Machine Learning by Stanford University on Coursera
2021	Genomic Data Science Specialization by Johns Hopkins University on Coursera
2020	Association Mapping: GWAS and Sequencing Data, 25th Summer Institute in Statistical Genetics, University of Washington, Seattle, WA
2020	Computational Pipeline for WGS Data, 25th Summer Institute in Statistical Genetics, University of Washington, Seattle, WA
2020	The Unix Workbench by Johns Hopkins University on Coursera
2019	American Language Institute Certified International Teaching Assistant
2019	Data Scientist with R Track on DataCamp
2017	R Programming by Johns Hopkins University on Coursera
2017	SAS Certified Advanced Programmer for SAS 9
2012	National Computer Rank Examination C Programming Level 2

HONORS & AWARDS

2020-2022	USC Graduate Research Assistantship
2019-2020	USC Graduate Teaching Assistantship
2018-2019	USC Graduate Student Fellowship
2015	Shanghai City Outstanding Graduate
2014	Jiang Hualiang (Dean) Scholarship First Prize
2013	Shanghai City Undergraduate Innovation Project
2013	ECUST Social Work Prize Level A
2013	ECUST Outstanding Student Leader
2012-2013	ECUST Outstanding Student
2012-2013	ECUST Integrated Curriculum First Prize
2012	ECUST Social Work Prize Level B
2012	Jiang Hualiang (Dean) Scholarship Third Prize
2012	The Fourth National Undergraduate Mathematical Competition Third Prize

PUBLICATIONS & PRESENTATIONS

13 peer-reviewed journal publications (4 first author), 3 conference presentations, 3 workshops teaching R programming, and 10 accepted conference abstracts. Full list available on [Google Scholar](#).

Publications

- Liu, T., **Xu, K.**, Pardeshi, A., Myint, S. S., Kang, A. Y., Morimoto, L. M., Lieber, M. R., Wiemels, J. L., Kogan, S. C., Metayer, C., de Smith, A. J. Early-life tobacco exposure is causally implicated in aberrant RAG-mediated recombination in childhood acute lymphoblastic leukemia. *Leukemia*, 2024.
- Feng, Q., **Xu, K.**, Shah, M., Li, S., Leavitt, A. D., Godley, L. A., de Smith, A. J., Wiemels, J. L.. Evaluation of the genetic basis of familial-associated early-onset hematologic cancers in an ancestral/ethnically diverse population. *Haematologica*, 2024.
- Wang, A., Xu, Y., Yu, Y., Nead, K. T., Kim, T., **Xu, K.**, Dadaev, T., Saunders, E., Sheng, X., Wan, P., . . . , Clonal hematopoiesis and risk of prostate cancer in large samples of European ancestry men. *Human Molecular Genetics*, 2023.
- Sanogo, F., **Xu, K.**, Cortessis, V., Weigensberg, M. J., Watanabe, R. M., Mind and Body-based Interventions Improve Glycemic Control in Patients with Type 2 Diabetes: A Systematic Review and Meta-Analysis. *Journal of Integrative and Complementary Medicine*, 2023.
- **Xu, K.**, Li, S., Pandey, P., Kang, A. Y., Morimoto, L. M., Mancuso, N., Ma, X., Metayer, C., Wiemels, J. L., de Smith, A. J., Investigating DNA methylation as a mediator of genetic risk in childhood acute lymphoblastic Leukemia. *Human Molecular Genetics*, 2022.
- **Xu, K.**, Li, S., Muskens, I. S., Elliott, N., Myint, S. S., Pandey, P., Hansen, H. M., Morimoto, L. M., Kang, A. Y., Ma, X., Metayer, C., Mueller, B. A., Roberts, I., Walsh, K. W., Horvath, S., Wiemels, J. L., de Smith, A. J., Accelerated Epigenetic Aging in Newborns with Down Syndrome. *Aging Cell*, 2022.
- Li, S., Sok, P., **Xu, K.**, Muskens, I. S., Elliott, N., Myint, S. S., Pandey, P., Hansen, H. M., Morimoto, L. M., Kang, A. Y., Metayer, C., Ma, X., Mueller, B. A., Roy, A., Roberts, I., Rabin, K. R., Brown, A. L., Lupo, P. J., Wiemels, J. L., de Smith, A. J., Epigenome-Wide Association Study of Acute Lymphoblastic Leukemia in Children with Down Syndrome. *Blood Adv.* 2022.
- Yu, F., Cato, L. D., Weng, C., Liggett, L. A., Jeon, S., **Xu, K.**, Chiang, C. W., Wiemels, J. L., Weissman, J. S., de Smith, A. J., Sankaran, V. G., Variant to function mapping at single-cell resolution through network propagation. *Nature Biotechnology*, 2022.
- **Xu, K.**, Feng, Q., Wiemels, J. L., de Smith, A. J., Disparities in acute lymphoblastic leukemia risk and survival across the lifespan in the United States of America. *Journal of Translational Genetics and*

Genomics, 2021.

- **Xu, K.**, Li, S., Whitehead, T. P., Pandey, P., Kang, A. Y., Morimoto, L. M., Kogan, S., Metayer, C., Wiemels, J. L., de Smith, A. J., Epigenetic Biomarkers of Prenatal Tobacco Smoke Exposure are Associated with Gene Deletions in Childhood Acute Lymphoblastic Leukemia. *Cancer Epidemiology and Prevention Biomarkers*, 2021. Featured in “Cancer Epidemiology, Biomarkers & Prevention Highlights: Selected Articles from This Issue.”
- Schott, S. L., Berkowitz, J., Dodged, S. E., Petersen, C. L., Saunders, C. H., Sobti, N. K., **Xu, K.**, Coylewright, M., Personalized, Electronic Health Record–Integrated Decision Aid for Stroke Prevention in Atrial Fibrillation: A Small Cluster Randomized Trial and Qualitative Analysis of Efficacy and Acceptability. *Circulation: Cardiovascular Quality and Outcomes*, 2021.
- Coylewright, M., O'Neill, E., Sherman, A., Gerling, M., Adam, K., **Xu, K.**, Grande, S. W., Dauerman, H. L., Dodged, S. E., Sobti, N. K., Saunders, C. H., Schott, S. L., Elwyn, G., Durand, M., The Learning Curve for Shared Decision-making in Symptomatic Aortic Stenosis. *JAMA cardiology*, 2020.
- Coylewright, M., Keevil, J. G., **Xu, K.**, Dodge, S., Frosch, D., Field, M. E., Pragmatic Study of Clinician Use of a Personalized Patient Decision Aid Integrated into the Electronic Health Record: An Eight-year Experience. *Telemedicine and e-Health*, 2019.

Abstracts

- Sanogo, F., **Xu, K.**, Cortessis, V., Weigensberg, M. J., Watanabe, R. M., Meta-analysis Shows Mind- and Body-Based Interventions Improve Glycemic Control in Patients with Type 2 Diabetes. Poster presentation at the American Diabetes Association's 82nd Scientific Sessions, 2022.
- **Xu, K.**, Wadé, N. B., Hwang, A., Conti, D. V., Salehi, M., Mack, T. M., Cortessis, V., van den Berg, D., de Smith, A. J., Cozen, W., Whole-exome sequencing in multiplex families to identify novel AYA classical Hodgkin lymphoma predisposition genes. Poster presentation at the 63rd American Society of Hematology (ASH) Annual Meeting and Exposition, 2021.
- Li, S., Sok, P., **Xu, K.**, Muskens, I. S., Elliott, N., Myint, S. S., Pandey, P., Hansen, H. M., Morimoto, L. M., Kang, A. Y., Metayer, C., Ma, X., Mueller, B. A., Roberts, I., Rabin, K. R., Brown, A. L., Lupo, P. J., Wiemels, J. L., de Smith, A. J., Epigenome-wide association study of acute lymphoblastic leukemia in children with Down syndrome. Oral presentation at the 63rd American Society of Hematology (ASH) Annual Meeting and Exposition, 2021.
- **Xu, K.**, Wadé, N. B., Hwang, A., Conti, D. V., Salehi, M., Mack, T. M., Cortessis, V., van den Berg, D., de Smith, A. J., Cozen, W., Whole-exome sequencing in multiplex families to identify novel AYA classical Hodgkin lymphoma predisposition genes. 15 minutes talk at the 2021 International Lymphoma Epidemiology Consortium Annual Meeting, 2021.
- **Xu, K.**, Li, S., Muskens, I. S., Elliott, N., Myint, S. S., Pandey, P., Ma, X., Metayer, C., Mueller, B. A., Walsh, K. W., Roberts, I., Horvath, S., Wiemels, J. L., de Smith, A. J., Accelerated epigenetic aging in newborns with Down syndrome. Poster presentation at the American Society of Human Genetics (ASHG) Virtual Meeting, 2021.
- Schott, S. L., **Xu, K.**, Berkowitz, J., Petersen, C., Saunders, C., Sobti, N., Coylewright, M., Timing of Electronic Health Record Integrated Decision Aid (IDEA) for Stroke Prevention in Atrial Fibrillation Matters. American College of Cardiology Scientific Sessions, 2019.
- Coylewright, M., Sherman, A., Grande, S. W., **Xu, K.**, Kirk, J., Dillon, G., O'Neill, E., Elwyn, G., Does a Telehealth “Virtual Consult” Including Referring Physicians, Specialist Physicians and Patients Increase Shared Decision Making for Patients with Heart Disease? Quality of Care and Outcomes Research Scientific Sessions, 2018.
- Coylewright, M., O'Neill, E., Sherman, A., Gerling, M., Adam, K., **Xu, K.**, Grande, S., Dauerman, H. L., Elwyn, G., A Learning Curve for Shared Decision Making: The Impact of Physician Experience on

Decision Aid Efficacy in Severe Aortic Stenosis. American College of Cardiology Scientific Sessions, 2018.

- Castaner, M. M., **Xu, K.**, Repka K., Gennetian L., Kennedy, J., Duch, H., A Multi-Dimensional Approach to Parent Engagement. Results from An Integrated School Readiness Intervention. Society for Research in Child Development, 2017 Biennial Meeting.
- **Xu, K.**, Castaner, M. M., Duch, H., A Multi-Dimensional Approach to Parent Engagement. Results from an Integrated School Readiness Intervention. Mailman School of Public Health. EPI Masters Student Day, 2016. Poster Presentation.

Conference Presentations

- Whole-exome sequencing in multiplex families to identify novel AYA classical Hodgkin lymphoma predisposition genes (Poster presentation). the 63rd American Society of Hematology (ASH) Annual Meeting and Exposition, Virtual. Dec. 2021.
- Accelerated epigenetic aging in newborns with Down syndrome (Poster presentation). the American Society of Human Genetics (ASHG) 2021 Meeting, Virtual. Oct. 2021.
- Whole-exome sequencing in multiplex families to identify novel AYA classical Hodgkin lymphoma predisposition genes (15-minute talk). the 2021 International Lymphoma Epidemiology Consortium Annual Meeting, Virtual. Sept. 2021.

Workshops

- Automatic Reports with RMarkdown. One hour workshop on RMarkdown for MPH/MS/PhD students at the Keck School of Medicine of USC, University of Southern California. Aug. 2020.
- Draw fractals from root finding iteration in R. Half hour workshop on drawing fractals from root finding iteration in R, LA R Users Group. April 2020.
- Building a Personal Website Using Blogdown. Two hour workshop on creating personal website using R package Blogdown, R-Ladies Pasadena. Nov. 2019.